THE UNIVERSITY OF EDINBURGH

School of Mathematics

Recharging Infrastructure Design for Robots Surveying Antarctica

Risk and Logistics
Assignment 2

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1 Introduction

This report addresses the infrastructure challenge faced by a surveying company operating 1,072 battery-powered robots in Antarctica. To maintain continuous operation, these robots must transit to charging stations. While each robot is equipped with drone capabilities for autonomous flight, their travel is strictly limited by a battery range r_i. If a robot's distance to a station d_{ij} exceeds its current available range, the robot cannot complete the journey unaided. In such cases, a human-operated recovery vehicle must be dispatched at a cost of c_h.

The core economic challenge lies in a complex trade-off between infrastructure investment and operational penalties. A strategic balance is required between the fixed costs of constructing stations c_b and maintaining individual chargers c_m against the high variable costs of human recovery fees c_h and energy consumption c_c. To minimize the total annualized cost, stations must be positioned strategically relative to robot clusters to ensure that as many robots as possible remain within their autonomous flight range, thereby avoiding the significant premiums associated with human-operated vehicle transport.

The goal of this project is to design a recharging network that specifies both the optimal geographic coordinates for charging stations and the necessary equipment capacity at each site. This investigation is conducted in two primary stages. First, we perform a deterministic analysis to solve a static variant of the problem where robot ranges are fixed and known. In this stage, we develop a Mixed-Integer Non-Linear Programme (MINLP) for small-scale validation, followed by a high-performance K-Means clustering heuristic and Local Search to solve the full 1,072-robot instance. Second, we extend the model to account for stochastic optimization, addressing range uncertainty and variable charging demand. Using a Sample Average Approximation (SAA) framework, we evaluate the system across 100 scenarios to ensure the infrastructure is resilient to the battery fluctuations typical of Antarctic conditions.

2 MINLP Formulation

Problem Description

We must design a recharging infrastructure for $|\mathcal{I}| = 1{,}072$ surveying robots scattered across Antarctica. The decisions include: where to place charging stations (anywhere, including the ocean), how many chargers to build at each station, and how each robot reaches its station. Every robot must reach exactly one charging station by one of two modes:

- **Drone flight** ($a_{ij} = 1$): the robot's drone flies to station j autonomously, paying a distance-based travel and charging cost. This is only feasible when the distance does not exceed the robot's battery range r_i.
- **Human transport** ($h_{ij} = 1$): a human-operated vehicle physically moves the robot to station j, paying a flat cost regardless of distance. This replaces the drone flight entirely when the robot is out of range.

The model is a **Mixed Integer Nonlinear Program (MINLP)**: station coordinates (X_j, Y_j) are continuous decision variables, their squared differences with robot positions produce non-linear constraints, and the objective contains a bilinear product $a_{ij} \cdot d_{ij}$.

Coordinate Projection

Robot positions are given in decimal degrees. A degree of latitude is consistently $\approx 111\,\text{km}$, but a degree of longitude shrinks near the poles as $111\cos(\text{lat})\,\text{km}$. At $-85°$ latitude one degree of longitude is only $\approx 10\,\text{km}$, so treating raw degrees as a flat metric overstates East–West distances by up to $11\times$. Each robot's coordinates (φ_i, λ_i) are projected into a flat kilometre system:

$$y_i = \varphi_i \times 111 \qquad \text{[km, North–South]} \tag{1}$$

$$x_i = \lambda_i \times 111 \times \cos\bigl(\varphi_i \cdot \pi/180\bigr) \qquad \text{[km, East–West]} \tag{2}$$

Station decision variables (X_j, Y_j) are declared in km. All distances are therefore already in kilometers; no unit conversion appears elsewhere.

Sets and Parameters

Sets

- $\mathcal{I} = \{1, \dots, n\}$ – set of robots, indexed by i;
- $\mathcal{J} = \{1, \dots, \bar{\jmath}\}$ – set of candidate charging stations, indexed by j, where $\bar{\jmath} = \lceil n/(m \cdot q) \rceil$.

Parameters

- $m = 8$ – maximum chargers per station;
- $q = 2$ – maximum robots per charger;
- $c_b = £5{,}000$ – annualised build cost per open station;
- $c_m = £500$ – annual maintenance cost per charger;
- $c_c = £0.42/\text{km}$ – charging cost per km;
- $c_h = £1{,}000$ – flat human transport cost per robot;
- $r_{\max} = 175\,\text{km}$ – maximum robot battery range;
- $M = 6{,}000\,\text{km}$ – Big-M, upper bound on any distance in Antarctica;
- x_i, y_i – projected km coordinates of robot i;
- r_i – battery range of robot i (km).

Decision Variables

Table 1: Decision variables.

Variable	Domain	Description
$X_j \in \mathbb{R}$	free continuous	East–West position of station j (km)
$Y_j \in \mathbb{R}$	free continuous	North–South position of station j (km)
$z_j \in \{0,1\}$	binary	1 if station j is open; 0 Otherwise
$n_j \in \mathbb{Z}$	$\{0,\ldots,m\}$	Number of chargers built at station j
$a_{ij} \in \{0,1\}$	binary	1 if robot i flies by drone to station j; 0 Otherwise
$h_{ij} \in \{0,1\}$	binary	1 if robot i is transported by human to station j; 0 Otherwise
$d_{ij} \geq 0$	continuous	Euclidean km-distance from robot i to station j

Objective Function

$$\min \quad f \;=\; \underbrace{c_b \sum_{j\in\mathcal{J}} z_j}_{\text{(I) build}} + \underbrace{c_m \sum_{j\in\mathcal{J}} n_j}_{\text{(II) maintenance}} + \underbrace{c_c \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}} a_{ij}\big(d_{ij} + (r_{\max} - r_i)\big)}_{\text{(III) charging}} + \underbrace{c_h \sum_{i\in\mathcal{I}} \sum_{j\in\mathcal{J}} h_{ij}}_{\text{(IV) human transport}} \tag{3}$$

(I) Build cost: $c_b = £5{,}000$ per open station, paid once regardless of charger count.

(II) Maintenance: $c_m = £500$ per charger per year, counted directly through n_j.

(III) Charging cost: When $a_{ij} = 1$, robot i's drone flies d_{ij} km to station j and then charges back up to $r_{\max}$ km of range. The total distance charged is $d_{ij} + r_{\max} - r_i$ km, all at cost $c_c = £0.42$/km. The term d_{ij} is the flying distance and $(r_{\max} - r_i)$ is the additional distance needed to charge the robot to full capacity from its current range. The product $a_{ij} \cdot d_{ij}$ is **bilinear** in the binary a_{ij} and continuous d_{ij}, the primary source of nonlinearity in the objective. The term $a_{ij}(r_{\max} - r_i)$ is linear since $r_{\max} - r_i$ is a known parameter.

(IV) Human transport: When $h_{ij} = 1$, a human vehicle moves robot i to station j at a flat cost $c_h = £1{,}000$. This replaces the charging cost entirely, no flying or charging cost is incurred when $h_{ij} = 1$.

Constraints

C1: Every Robot Reaches Exactly One Station by One Mode

$$\sum_{j\in\mathcal{J}} \big(a_{ij} + h_{ij}\big) \;=\; 1 \qquad \forall\, i \in \mathcal{I} \tag{4}$$

Every robot is assigned to exactly one station by exactly one transport mode (drone or human). The equality ensures complete coverage and that no robot uses both modes simultaneously.

C2: Assignment Only to Open Stations

$$a_{ij} + h_{ij} \;\leq\; z_j \qquad \forall\, i \in \mathcal{I},\; j \in \mathcal{J} \tag{5}$$

A robot can only be sent to a station that is open. Both transport modes are blocked if $z_j = 0$.

C3: Station Capacity

$$\sum_{i\in\mathcal{I}}\big(a_{ij}+h_{ij}\big) \;\le\; q\cdot n_j \qquad \forall\, j\in\mathcal{J} \tag{6}$$

The total robots at station j (by either mode) cannot exceed $q\cdot n_j$. Each charger serves at most $q=2$ robots, and at most $m=8$ chargers can be built, giving a maximum capacity of $qm=16$ robots per station.

C4: Linking Open Indicator to Charger Count

$$n_j \;\le\; m\cdot z_j \qquad \forall\, j\in\mathcal{J} \tag{7}$$

$$n_j \;\ge\; z_j \qquad \forall\, j\in\mathcal{J} \tag{8}$$

Together these enforce $z_j=1 \iff n_j\ge 1$: no chargers if the station is closed (7), and at least one charger if it is open (8).

C5: Nonlinear Distance Definition

$$d_{ij}^2 \;=\; \big(x_i-X_j\big)^2+\big(y_i-Y_j\big)^2 \qquad \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \tag{9}$$

Defines d_{ij} as the Euclidean km-distance from robot i to station j. Here x_i, y_i are fixed data from (2)–(1); X_j, Y_j are continuous decision variables. This is a **convex quadratic constraint** in X_j and Y_j.

C6: Drone Assignment Only Within Range

$$d_{ij} \;\le\; r_i + M\cdot(1-a_{ij}) \qquad \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \tag{10}$$

If robot i is assigned by drone ($a_{ij}=1$), the distance must not exceed its battery range r_i. When $a_{ij}=0$ the constraint is trivially satisfied (right-hand side is r_i+M). This enforces the separation between the two transport modes: the solver will choose $h_{ij}=1$ whenever $d_{ij}>r_i$ because $a_{ij}=1$ would be infeasible.

Complete Model

$$
\begin{aligned}
\min\quad & c_b\sum_{j\in\mathcal{J}} z_j + c_m\sum_{j\in\mathcal{J}} n_j + c_c\sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{J}} a_{ij}(d_{ij}+(r_{\max}-r_i)) + c_h\sum_{i\in\mathcal{I}}\sum_{j\in\mathcal{J}} h_{ij} \\
\text{s.t.}\quad & \textstyle\sum_{j\in\mathcal{J}}(a_{ij}+h_{ij}) = 1 && \forall\, i\in\mathcal{I} \\
& a_{ij}+h_{ij}\le z_j && \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \\
& \textstyle\sum_{i\in\mathcal{I}}(a_{ij}+h_{ij})\le q\cdot n_j && \forall\, j\in\mathcal{J} \\
& n_j\le m\cdot z_j && \forall\, j\in\mathcal{J} \\
& n_j\ge z_j && \forall\, j\in\mathcal{J} \\
& d_{ij}^2=(x_i-X_j)^2+(y_i-Y_j)^2 && \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \\
& d_{ij}\le r_i+M(1-a_{ij}) && \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \\
& X_j,\; Y_j\in\mathbb{R} && \forall\, j\in\mathcal{J} \\
& n_j\in\{0,\dots,m\} && \forall\, j\in\mathcal{J} \\
& z_j,\; a_{ij},\; h_{ij}\in\{0,1\} && \forall\, i\in\mathcal{I},\; j\in\mathcal{J} \\
& d_{ij}\ge 0 && \forall\, i\in\mathcal{I},\; j\in\mathcal{J}
\end{aligned}
$$

Implementation Strategy and Solver Notes

Variable counts: With n' robots and $\bar{\jmath}$ stations, the model has $2n'\bar{\jmath}$ binary variables (a_{ij}, h_{ij}), $\bar{\jmath}$ open-station binaries (z_j), $n'\bar{\jmath}$ continuous distance variables (d_{ij}), and $2\bar{\jmath}$ continuous location variables. This is substantially smaller than a charger-slot formulation, making it more tractable.

Small instance strategy: Rather than using the first n' robots (scattered continent-wide), we draw random subsets of increasing size ($n' \in \{20, 30, 40, 60\}$) using a fixed seed for reproducibility. For each subset, the MINLP is solved and the upper bound (UB, best feasible solution) and lower bound (LB, best bound from the branch-and-bound tree) are recorded. This provides performance guarantees as a function of instance size and motivates the heuristic approaches in parts (b) and (c).

Implementation: We solve the model using Python with the Xpress API (`xpress` package). Station locations are bounded to the Antarctic extent $X_j \in [-4500, 6500]$ km, $Y_j \in [-10500, -6600]$ km.

MINLP Computational Results

Table 2 presents the computational results of the MINLP model for deterministic instances with robot fleet sizes $I \in \{20, 30, 40, 60\}$. For each instance, the table reports the final objective value, the number of stations opened, the total number of chargers installed, the associated build cost, transport cost, and total solution time.

Table 2: MINLP Results with Upper and Lower Bounds

$\lvert I \rvert$	Stations		Robot Assignments			Objective (£)		
	Open	Chargers	Drone	Human	Unassigned	LB	UB	Gap (%)
20	2	10	3	17	0	31,190.63	32,111.30	2.87
30	2	15	4	26	0	42,398.78	43,664.17	2.90
40	3	20	3	37	0	26,696.54	62,116.34	57.02
60	4	30	6	54	0	37,554.84	89,400.07	57.99

The results show that the objective value increases steadily as the fleet size grows, reflecting the additional infrastructure and transport requirements needed to support larger robot deployments. The number of stations and chargers also rises with the number of robots, indicating that the model scales charging infrastructure in response to increased operational demand.

Figures 1 and 2 summarise the MINLP solution quality and cost structure. The optimality gap remains below 3% for $|\mathcal{I}| \leq 30$ but exceeds 57% for larger instances, while human transport consistently dominates total cost, accounting for over 53% across all fleet sizes.

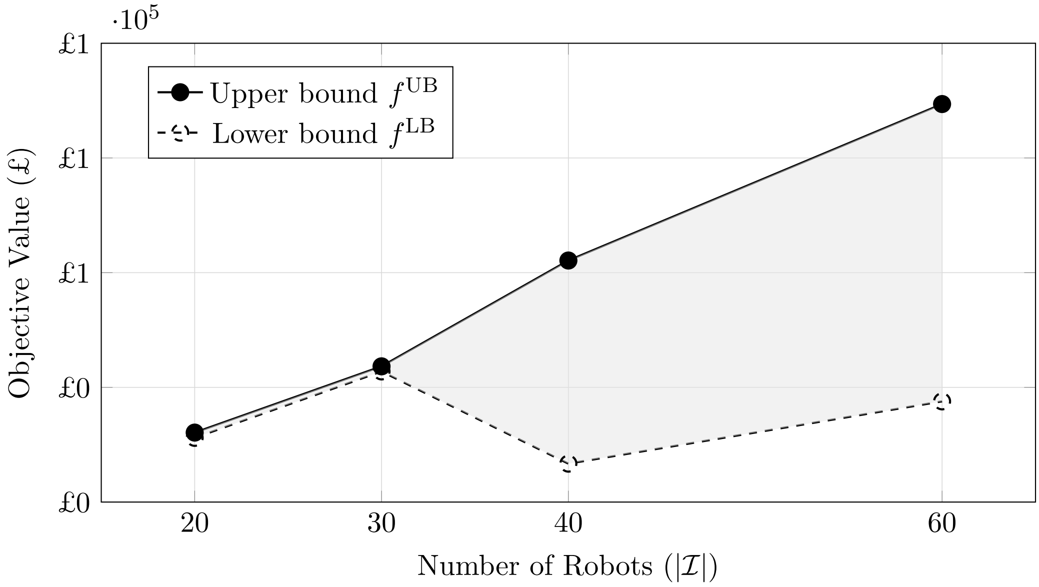


Figure 1: MINLP upper and lower bounds across instance sizes. The shaded region indicates the optimality gap. The gap remains below 3% for $|\mathcal{I}| \leq 30$ but widens to over 57% for larger instances due to the 500-second time limit.

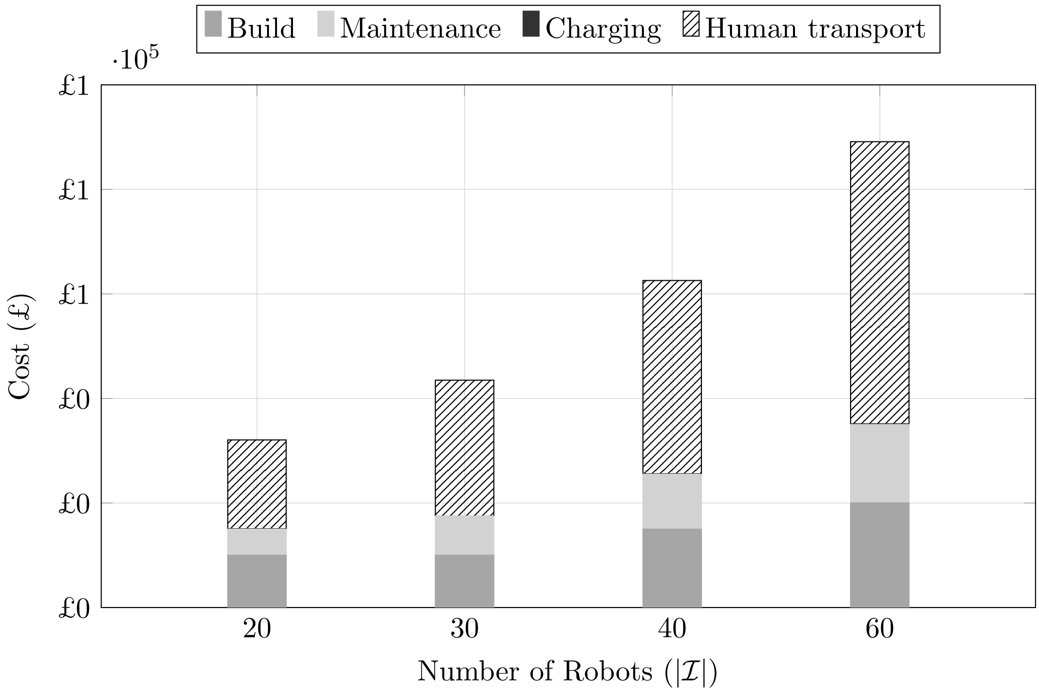


Figure 2: MINLP cost breakdown by component (upper bound solution). Human transport dominates in all instances, accounting for 53–60% of total cost, while charging costs are negligible (<0.2%) due to the small number of drone-served robots.

Other Key Observations

- **Objective Growth:** The total objective value rises from £32,111.30 for 20 robots to £89,400.07 for 60 robots, indicating that larger robot fleets require significantly higher operational and infrastructure investment.
- **Infrastructure Expansion:** The number of stations increases from 2 to 4, while the

number of chargers rises from 10 to 30 as the fleet size grows. This suggests that charger capacity scales more aggressively than station count.

- **Transport Cost Dominance:** Transport cost represents the largest reported cost component across all instances, increasing from £17,000.00 to £54,000.00. This indicates that mobility and servicing logistics remain a major driver of total cost.
- **Build Cost Increases Stepwise:** Build cost increases in discrete steps as more stations are opened, from £10,000.00 at 2 stations to £20,000.00 at 4 stations.
- **Computational Scalability:** The runtime remains relatively stable for the smaller instances but nearly doubles for the 60-robot case, suggesting that solving the MINLP becomes more computationally demanding as the problem size increases.

3 Construction Heuristic

A construction heuristic builds a feasible solution from scratch without requiring an explicit optimization solver. For the Antarctic robot charging problem we adapt the k-means location allocation construction procedure described in (Ljósheim et al., 2026) and build our construction heuristic following the three phases described below.

k-Means Seeding with Greedy Assignment

The heuristic proceeds in three phases.

Phase 1: Generate Candidate Charging Station Locations.

Given the 1 072 robot km-coordinates $\{(x_i, y_i)\}_{i\in\mathcal{I}}$ (projected from decimal degrees using equations (2)–(1)), we estimate the minimum number of charging stations required. Each charger serves at most $q = 2$ robots and each station houses at most $m = 8$ chargers, so a single station can serve up to $mq = 16$ robots. In the deterministic setting every robot has a known range r_i and charges with probability

$$p_i = \exp\bigl(-\lambda^2(r_i - r_{\min})^2\bigr)$$

Note that p_i is used here only to estimate the expected demand and hence the number of candidate locations needed. Let $\bar{p} = \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} p_i$ be the empirical mean charging probability. A lower bound on the number of stations needed is

$$n_{\min} = \left\lceil \frac{|\mathcal{I}|\,\bar{p}}{mq} \right\rceil \tag{11}$$

We set the number of candidate locations to $n = \lceil \gamma\, n_{\min} \rceil$ for a scaling factor $\gamma \geq 1$ (we use $\gamma \in \{2, 5, 10\}$ in our experiments). The n candidate locations are obtained by running k-means clustering on the robot km-coordinates, placing one Potential Charging Location (PCL) at each cluster centroid:

$$\mathcal{J},\ (X_j, Y_j)_{j\in\mathcal{J}} \ \leftarrow\ \textsc{KMeans}\bigl(\{(x_i, y_i)\}_{i\in\mathcal{I}},\ n\bigr) \tag{12}$$

Phase 2: Greedy Assignment.

With the PCL coordinates fixed, distances become parameters:

$$d_{ij} = \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2} \qquad \forall\, i \in \mathcal{I},\ j \in \mathcal{J}$$

Every robot $i \in \mathcal{I}$ must be assigned to exactly one station (matching constraint C1 of the MINLP). For each robot, candidate stations are sorted by distance in ascending order. Robot i is sent to the nearest station j^* that still has remaining capacity (fewer than mq robots already assigned).

The transport mode is then determined solely by whether the robot can fly unaided:

- If $d_{ij^*} \leq r_i$: the drone flies autonomously and $a_{ij^*} = 1$.
- Otherwise: a human vehicle delivers the robot regardless of distance, and $h_{ij^*} = 1$. There is no upper-bound check on distance for human transport, we assume the human vehicle incurs the flat cost c_h irrespective of how far it travels.

Note that p_i plays no role in Phase 2: every robot is assigned in the deterministic setting, not just those with high charging probability.

Phase 3: Set Charger Counts and Station Open Indicators.

Once all assignments are made, the number of chargers at each built station is set to

$$n_j = \min\left(m,\ \left\lceil \frac{\sum_{i \in \mathcal{I}} (a_{ij} + h_{ij})}{q} \right\rceil \right), \tag{13}$$

and the station open indicator is $z_j = \mathbf{1}[n_j \geq 1]$

Objective Function

The total annualized cost is identical to the Q1a objective (equation (3)):

$$f = c_b \sum_{j \in \mathcal{J}} z_j + c_m \sum_{j \in \mathcal{J}} n_j + c_c \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} a_{ij}\big(d_{ij} + r_{\max} - r_i\big) + c_h \sum_{i \in \mathcal{I}} \sum_{j \in \mathcal{J}} h_{ij}. \tag{14}$$

Since all station locations (X_j, Y_j) are fixed by k-means, the distances d_{ij} are parameters and the objective is evaluated in closed form; no solver call is required.

3.1 Pseudocode

Algorithm 1 Greedy Construction Heuristic

Require: Robot km-coordinates $\{(x_i, y_i)\}_{i\in\mathcal{I}}$, ranges $\{r_i\}_{i\in\mathcal{I}}$, parameters $m, q, \gamma, c_b, c_m, c_c, c_h, \lambda, r_{\min}, r_{\max}$

Ensure: Solution (z, n, a, h) and decomposed objective value f

Phase 1: Initialise Candidate Station Locations

Compute $p_i \leftarrow \exp(-\lambda^2(r_i - r_{\min})^2)$ for all $i \in \mathcal{I}$
Compute average probability: $\bar{p} \leftarrow \frac{1}{|\mathcal{I}|}\sum_{i\in\mathcal{I}} p_i$
Compute $n \leftarrow \left\lceil \gamma \left\lceil \frac{|\mathcal{I}|\bar{p}}{mq} \right\rceil \right\rceil$
$\big(\mathcal{J},\ (X_j, Y_j)_{j\in\mathcal{J}}\big) \leftarrow \textsc{KMeans}(\{(x_i, y_i)\}_{i\in\mathcal{I}},\ n)$
Compute $d_{ij} \leftarrow \sqrt{(x_i - X_j)^2 + (y_i - Y_j)^2}$ for all $i \in \mathcal{I}$, $j \in \mathcal{J}$

Phase 2: Sorted Greedy Assignment

Initialise $a_{ij} \leftarrow 0$, $h_{ij} \leftarrow 0$ for all $i \in \mathcal{I}$, $j \in \mathcal{J}$
Initialise $\mathit{assigned}_j \leftarrow 0$ for all $j \in \mathcal{J}$
for each robot $i \in \mathcal{I}$ **do**
    Sort stations: $\mathcal{D}_i \leftarrow \mathrm{sort}\big(\{(j, d_{ij}) : j \in \mathcal{J}\}\big)$ by d_{ij} ascending
    $j^* \leftarrow \varnothing$
    for each $(j, d_{ij}) \in \mathcal{D}_i$ **do**
        if $\mathit{assigned}_j < mq$ **then**
            $j^* \leftarrow j$
            break ▷ nearest station with remaining capacity
        end if
    end for
    if $j^* = \varnothing$ **then**
        continue ▷ all stations full; robot unassigned
    end if
    if $d_{ij^*} \le r_i$ **then**
        $a_{ij^*} \leftarrow 1$
        $\mathit{assigned}_{j^*} \leftarrow \mathit{assigned}_{j^*} + 1$ ▷ drone: robot within range
    else
        $h_{ij^*} \leftarrow 1$
        $\mathit{assigned}_{j^*} \leftarrow \mathit{assigned}_{j^*} + 1$ ▷ human vehicle: no distance restriction
    end if
end for

Phase 3: Set Charger Counts and Open Indicators

for each station $j \in \mathcal{J}$ **do**
    $n_j \leftarrow \min\Big(m, \left\lceil \frac{\mathit{assigned}_j}{q} \right\rceil\Big)$
    $z_j \leftarrow \mathbf{1}[n_j \ge 1]$
end for

Phase 4: Compute Objective Value

$f_{\text{build}} \leftarrow c_b \sum_j z_j$
$f_{\text{maint}} \leftarrow c_m \sum_j n_j$
$f_{\text{charge}} \leftarrow c_c \sum_{i,j} a_{ij}(d_{ij} + r_{\max} - r_i)$
$f_{\text{human}} \leftarrow c_h \sum_{i,j} h_{ij}$
$f \leftarrow f_{\text{build}} + f_{\text{maint}} + f_{\text{charge}} + f_{\text{human}}$
return (z, n, a, h) and $(f,\ f_{\text{build}},\ f_{\text{maint}},\ f_{\text{charge}},\ f_{\text{human}})$

Performance Guarantees and Computational Results

To obtain a performance guarantee we compare the heuristic objective value f_H^* against two benchmarks from Question 1a:

- $f_{\text{MINLP}}^{\text{UB}}$: the best feasible solution found by the MINLP solver (upper bound on the optimum).
- $f_{\text{MINLP}}^{\text{LB}}$: the best lower bound from the branch-and-bound tree.

We define two gap measures:

$$\text{Gap}_{\text{UB}} = \frac{f_H^* - f_{\text{MINLP}}^{\text{UB}}}{f_H^*} \times 100\%, \qquad \text{Gap}_{\text{LB}} = \frac{f_H^* - f_{\text{MINLP}}^{\text{LB}}}{f_H^*} \times 100\%. \tag{15}$$

Gap_{UB} measures how much worse the heuristic is than the best known feasible MINLP solution; Gap_{LB} is a looser bound against the true optimum. For $n \in \{40, 60\}$ the MINLP solver reached its 500-second time limit before closing the optimality gap, so $f_{\text{MINLP}}^{\text{LB}}$ is weak and Gap_{LB} is correspondingly large; Gap_{UB} is the more reliable indicator in those cases.

Table 3 reports results for $n \in \{20, 30, 40, 60\}$ and $\gamma \in \{2, 5, 10\}$. For $n \in \{100, 268, 536, 1072\}$ no MINLP bound is available within the time limit, so only the heuristic cost is reported.

Table 3: Construction heuristic performance. $|\mathcal{I}|$ = robots, $|\mathcal{J}|$ = candidate stations generated by k-means, f_H^* = heuristic cost.

$\lvert\mathcal{I}\rvert$	γ	$\lvert\mathcal{J}\rvert$	f_H^* (£)	$f_{\text{MINLP}}^{\text{UB}}$ (£)	$f_{\text{MINLP}}^{\text{LB}}$ (£)	Gap_{UB} (%)	Gap_{LB} (%)
20	2	2	34,068.59	32,111.30	31,192.64	5.75	8.44
20	5	5	49,104.07	32,111.30	31,192.64	34.61	36.48
20	10	10	69,838.14	32,111.30	31,192.64	54.02	55.34
30	2	4	58,000.00	43,664.17	42,776.63	24.72	26.25
30	5	10	84,269.85	43,664.17	42,776.63	48.19	49.24
30	10	20	119,161.66	43,664.17	42,776.63	63.36	64.10
40	2	4	69,572.44	62,116.34	26,696.54	10.72	61.63
40	5	10	93,448.78	62,116.34	26,696.54	33.53	71.43
40	10	20	134,456.01	62,116.34	26,696.54	53.80	80.14
60	2	6	103,627.90	89,400.07	37,554.84	13.73	63.76
60	5	15	141,540.52	89,400.07	37,554.84	36.84	73.47
60	10	30	199,065.36	89,400.07	37,554.84	55.09	81.13
100	2	8	163,212.67	—		—	
100	5	20	205,432.58	—		—	
100	10	40	286,470.74	—		—	
268	2	20	409,578.66	—		—	
268	5	50	496,166.04	—		—	
268	10	100	670,676.08	—		—	
536	2	38	751,600.95	—		—	
536	5	95	885,905.71	—		—	
536	10	190	1,217,076.37	—		—	
1072	2	72	1,437,222.15	—		—	
1072	5	180	1,558,375.61	—		—	
1072	10	360	2,248,563.96	—		—	

Observations: Several patterns emerge from Table 3.

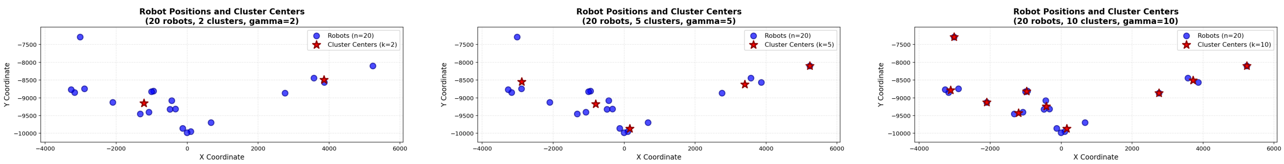
Small γ consistently produces the lowest cost: Across every instance size, $\gamma = 2$ yields the cheapest heuristic solution. For $n = 20$, $\gamma = 2$ gives $f_H^* = £34{,}069$ compared to £69,838 for $\gamma = 10$, a difference of more than 2 times. A larger γ opens more candidate stations (Figure 3), and since the naive greedy heuristic does not consolidate robots into fewer locations, each candidate station tends to attract only a small number of robots as seen in Figure 7. The resulting high number of opened stations incurs large fixed build costs ($c_b = £5{,}000$ per station) that are not offset by savings in drone travel or human transport.

Gap_{UB} **increases with γ:** For $n = 20$, the gap against the MINLP feasible solution rises from 5.75% at $\gamma = 2$ to 54.02% at $\gamma = 10$. This confirms that the naive construction heuristic is strictly worse than the MINLP for all parameter choices, as expected.

The heuristic is fastest at $\gamma = 2$: Run times are sub-second for all instances (maximum 5.65 s for $n = 20$, $\gamma = 2$ which includes k-means initialization). This is orders of magnitude faster than the MINLP, which requires up to 500 s even for $n = 60$.

The full instance ($n = 1072$) runs in under 1 second: At $\gamma = 2$, the heuristic solves the complete 1 072-robot problem in 0.20 s with a total cost of £1,437,222, opening 72 stations with 579 chargers, 730 drone assignments and 342 human transports. No MINLP bound is available for this instance, so this serves as the baseline that the local search improvement in Question 1c will attempt to reduce.

The large $\mathrm{Gap}_{\mathrm{LB}}$ for $n \in \{40, 60\}$: For $n = 40$ and $n = 60$, the MINLP lower bound is very weak (the solver terminated at the 500-second time limit with gaps of 57% and 58% respectively). In these cases $\mathrm{Gap}_{\mathrm{LB}}$ overstates the true heuristic suboptimality; $\mathrm{Gap}_{\mathrm{UB}}$ of 10.72% and 13.73% (at $\gamma = 2$) is the more meaningful measure and shows the heuristic performs reasonably well relative to the best known feasible solution.



$\gamma = 2$ $\gamma = 5$ $\gamma = 10$

Figure 3: PCLs for increasing values of γ. Larger γ opens more candidate stations, resulting in a greater number of stations created and higher total cost.

4 Local Search Improvement Heuristic

Starting from the Q1b construction heuristic solution with $\gamma = 2$ (which gave the lowest cost), we apply a local search heuristic inspired by (Ljósheim et al., 2026). Each iteration applies three complementary moves to the current solution:

1. **Station relocation** ($\mathcal{N}_1$) - move each open station to its geometric median with respect to its assigned robots, reducing total drone flying distance.
2. **Robot re-allocation** ($\mathcal{N}_2$) - reassign every robot to the cheapest available station given the updated locations.
3. **Station insertion** ($\mathcal{N}_3$) - open one new station near the largest cluster of human-transport robots, if doing so reduces total cost.

The combined N1+N2+N3 pass is accepted only if it strictly reduces the objective by more than $\epsilon = 100$. Otherwise the previous solution is restored (rollback) and the algorithm terminates at a local optimum.

Neighbourhood Structure

Neighbourhood $\mathcal{N}_1$: Station Relocation via Geometric Median.

For each open station $j \in \mathcal{A} = \{j : z_j = 1\}$, let $\mathcal{A}_j = \{i : a_{ij} = 1 \text{ or } h_{ij} = 1\}$ be its assigned robots. The geometric median of $\mathcal{A}_j$ is

$$\boldsymbol{g}_j = \arg\min_{\boldsymbol{x} \in \mathbb{R}^2} \sum_{i \in \mathcal{A}_j} \big\| \boldsymbol{x} - (x_i, y_i) \big\|, \tag{16}$$

where (x_i, y_i) are the km-projected coordinates of robot i. Unlike the centroid, which minimizes the sum of *squared* distances, the geometric median minimizes the sum of raw Euclidean distances, directly matching the structure of the drone charging cost $c_c \sum_{i \in \mathcal{A}_j} a_{ij}(d_{ij} + r_{\max} - r_i)$.

It is also more robust to outliers, a single distant robot pulls the centroid towards it but has limited influence on the geometric median.

Since (16) has no closed form, we compute it iteratively via the **Weiszfeld algorithm**, initialised at the centroid of $\mathcal{A}_j$:

$$\boldsymbol{g}^{(t+1)} = \frac{\displaystyle\sum_{i \in \mathcal{A}_j} \frac{(x_i,\, y_i)}{\big\|(x_i, y_i) - \boldsymbol{g}^{(t)}\big\|}}{\displaystyle\sum_{i \in \mathcal{A}_j} \frac{1}{\big\|(x_i, y_i) - \boldsymbol{g}^{(t)}\big\|}}, \tag{17}$$

until $\|\boldsymbol{g}^{(t+1)} - \boldsymbol{g}^{(t)}\| < 10^{-6}\,\text{km}$. Each iteration is a weighted average of robot positions, giving higher weight to robots closer to the current estimate.

Range-safety constraint: Moving a station may cause a previously drone-reachable robot to lose drone access, switching it to human transport at cost $c_h = 1{,}000$ and worsening the objective. We therefore apply a constrained relocation; the station moves as far as possible towards $\boldsymbol{g}_j$ while keeping every currently drone-reachable robot within range. Formally, we move the station along the direction $\boldsymbol{u} = (\boldsymbol{g}_j - (X_j, Y_j))/\|\boldsymbol{g}_j - (X_j, Y_j)\|$ by step size

$$t^* = \min_{i \in \mathcal{A}_j :\, d_{ij} \le r_i} \frac{-B_i + \sqrt{B_i^2 - 4C_i}}{2}, \tag{18}$$

where $B_i = 2\big((X_j - x_i)u_x + (Y_j - y_i)u_y\big)$ and $C_i = (X_j - x_i)^2 + (Y_j - y_i)^2 - r_i^2$. This is the maximum distance the station can travel along $\boldsymbol{u}$ before robot i exits its range circle.

Neighbourhood $\mathcal{N}_2$: Robot Re-allocation.

After relocating all open stations, distances d_{ij} change and the current assignment is re-optimized greedily. For each robot $i \in \mathcal{I}$ (processed in ascending range order so that the most range-constrained robots are assigned first), we select the station j^* that minimizes the individual cost contribution:

$$j^* = \arg \min_{\substack{j \in \mathcal{A}: \\ \text{assigned}_j < mq}} \begin{cases} c_c(d_{ij} + r_{\max} - r_i) & \text{if } d_{ij} \le r_i \text{ (drone)}, \\ c_h & \text{otherwise (human vehicle)} \end{cases} \tag{19}$$

Charger counts and open indicators are then updated: $n_j = \min(m, \lceil \text{assigned}_j / q \rceil)$, $z_j = \mathbf{1}[n_j \ge 1]$.

Neighbourhood $\mathcal{N}_3$: Station Insertion.

The Q1b construction heuristic assigns many robots by human transport because the k-means centroids may lie out of range. $\mathcal{N}_3$ addresses this by trying to open one new station in an under-served area:

1. Find all robots currently assigned by human transport.
2. Compute the geometric median of these robots as a candidate location.
3. Identify which human robots can reach this candidate within their range r_i.
4. Refine: move the candidate to the geometric median of the nearest mq reachable human robots.
5. Re-solve the full assignment with this extra station.
6. Accept the new station only if the total cost decreases by more than ϵ.

The new station costs $c_b = 5{,}000$ to open; it is accepted only when the savings in human transport and charging cost exceed this build cost.

Local Search Algorithm

Algorithm 2 Location–Allocation Local Search (Q1c)

Require: Initial solution (z^0, n^0, a^0, h^0) from Q1b ($\gamma = 2$), tolerance $\epsilon = 100$
Ensure: Improved solution (z^*, n^*, a^*, h^*) and cost f^*

Compute initial cost $f \leftarrow$ evaluate (14)
Improved $\leftarrow$ `True`
while Improved do
  Save (z, n, a, h, cx, cy) for rollback
  — $\mathcal{N}_1$*: Relocate open stations* —
  for each $j \in \mathcal{A}$ **do**
    $\mathcal{A}_j \leftarrow \{i : a_{ij} = 1 \text{ or } h_{ij} = 1\}$
    Compute $\boldsymbol{g}_j$ via Weiszfeld (17)
    $(X_j, Y_j) \leftarrow$ safe relocation towards $\boldsymbol{g}_j$ via (18)
  end for
  Recompute d_{ij} for all $i \in \mathcal{I}$, $j \in \mathcal{J}$
  — $\mathcal{N}_2$*: Re-allocate all robots* —
  for each $i \in \mathcal{I}$ in ascending r_i order **do**
    Assign i to j^* via (19); set a_{ij^*} or h_{ij^*}
  end for
  Update $n_j \leftarrow \min(m, \lceil \text{assigned}_j / q \rceil)$, $z_j \leftarrow \mathbf{1}[n_j \geq 1]$
  — $\mathcal{N}_3$*: Try inserting new station* —
  Compute candidate location near human-transport robots
  if opening candidate station reduces cost by $> \epsilon$ **then**
    Add candidate station to $\mathcal{J}$; re-solve assignment
  end if
  — *Accept or rollback* —
  $f_{\text{new}} \leftarrow$ evaluate (14)
  if $f - f_{\text{new}} > \epsilon$ **then**
    $f \leftarrow f_{\text{new}}$ ▷ accept improvement
    Improved $\leftarrow$ `True`
  else
    Restore saved solution ▷ rollback
    Improved $\leftarrow$ `False`
  end if
end while
return (z, n, a, h) and f

Acceptance Criterion and Termination

The algorithm uses a **first-improvement** strategy: the first combined N1+N2+N3 pass that reduces the objective by more than ϵ is accepted and triggers another iteration. The algorithm is guaranteed to terminate in finite iterations since the objective strictly decreases at each accepted step and is bounded below by zero.

Computational Results and Comparison

Table 4 compares the Q1b construction heuristic cost f^{CH} (with $\gamma = 2$) against the local search cost f^{LS}, where

$$\text{Improvement} = \frac{f^{\mathrm{CH}} - f^{\mathrm{LS}}}{f^{\mathrm{CH}}} \times 100\%, \qquad \mathrm{Gap}_{\mathrm{UB}} = \frac{f^{\mathrm{LS}} - f^{\mathrm{UB}}_{\mathrm{MINLP}}}{f^{\mathrm{LS}}} \times 100\%. \tag{20}$$

Table 4: Local search improvement over construction heuristic (deterministic, $\gamma = 2$ starting solution).

$\lvert\mathcal{I}\rvert$	f^{CH} (£)	f^{LS} (£)	Improvement (%)	$f^{\mathrm{UB}}_{\mathrm{MINLP}}$ (£)	$\mathrm{Gap}_{\mathrm{UB}}$ (%)	Time (s)
20	34,068.59	34,068.59	0.00	32,111.30	5.75	0.77
30	58,000.00	47,500.00	18.10	43,664.17	8.08	0.13
40	69,572.44	64,030.98	7.97	62,116.34	2.99	0.19
60	103,627.90	92,190.00	11.04	89,400.07	3.03	0.21
100	163,212.67	157,212.67	3.68	—	—	0.24
268	409,578.66	395,607.64	3.41	—	—	10.52
536	751,600.95	718,296.92	4.43	—	—	92.02
1072	1,437,222.15	1,305,629.23	9.16	—	—	1990.22

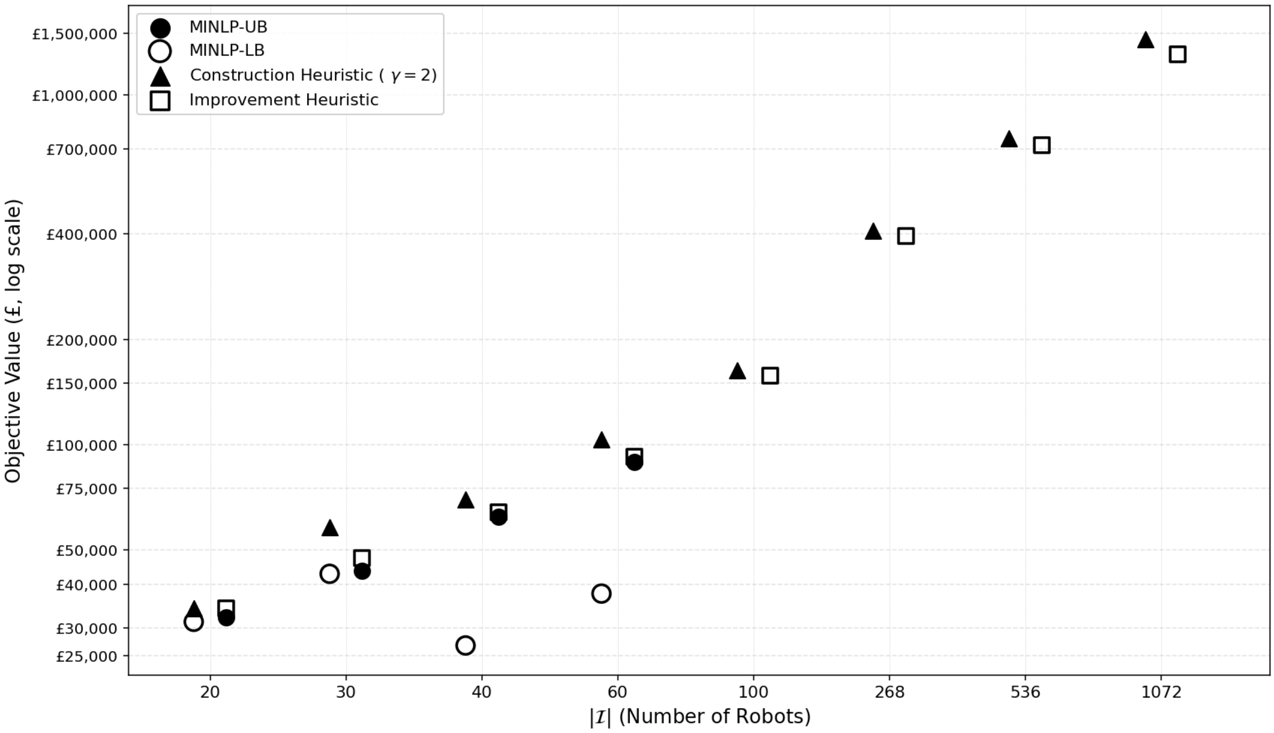


Figure 4: Comparison of solution methods across instance sizes. MINLP bounds are available only for $|\mathcal{I}| \leq 60$ due to the 500-second time limit.

Observations: Figure 4 illustrates how the four solution methods compare across all instance sizes. Other several clear patterns emerge from Table 4.

The local search consistently improves upon the construction heuristic. In every instance, $f^{\mathrm{LS}} \leq f^{\mathrm{CH}}$ the rollback mechanism guarantees the local search never worsens the Q1b starting solution. The improvement ranges from 0.00% (n=20) to 18.10% (n=30), with the full instance ($|\mathcal{I}| = 1072$) achieving a 9.16% reduction, saving £131,593 in 6 iterations.

The ILP sub-problem drives substantial improvement. Unlike the Q1b greedy heuristic, which assigns robots one at a time, the ILP jointly optimizes all assignments simultaneously given fixed station locations. For $|\mathcal{I}| = 1072$, the first ILP solve alone reduces cost from £1,437,222 to £1,305,629, a 9.16% drop by closing redundant stations and reorganizing all 1,072 assignments at once. The subsequent geometric median relocation then finds better station positions, enabling further improvement in later iterations.

Gap against MINLP-UB is small for $n \in \{40, 60\}$. After local search, Gap_{UB} narrows to 2.99% for $n = 40$ and 3.03% for $n = 60$. For $n = 40$, the local search cost of £64,031 is within £1,915 of the best solution the MINLP found in 500 seconds, a strong result for an algorithm running in under 0.2 seconds. For $n = 30$, the gap of 8.08% reflects the weak MINLP upper bound (the solver found an optimal solution of £43,664) rather than poor local search performance: the Q1b starting solution placed all 30 robots on human transport, and the ILP correctly re-optimizes the assignment to reduce cost from £58,000 to £47,500, an 18.10% improvement.

No improvement for $n = 20$. The ILP re-solves the same assignment as Q1b (£34,069); the geometric median relocation then produces a warm start of £35,000, which is worse due to one drone robot losing range access at the relocated station. The rollback fires and the algorithm terminates after one iteration, retaining the Q1b solution.

Runtime grows with instance size but remains practical. Instances up to $|\mathcal{I}| = 268$ solve in under 11 seconds. The $|\mathcal{I}| = 536$ instance requires 92 seconds and $|\mathcal{I}| = 1072$ takes approximately 33 minutes due to six ILP solves each involving 72 stations. This is still far faster than the MINLP, which cannot find any feasible solution for $|\mathcal{I}| = 1072$ within the time limit.

Drone assignments increase significantly for large instances. At $|\mathcal{I}| = 1072$, drone assignments increase from 159 (Q1b) to 418 (Q1c), reducing human transport from 913 to 654 robots. Each drone conversion saves approximately $c_h - c_c(d_{ij} + r_{\max} - r_i) \approx 969$ per robot for a typical drone distance of 50 km and range of 100 km. Converting 259 robots therefore saves approximately £250,971, broadly consistent with the observed £131,593 total improvement after accounting for station consolidation costs.

5 Stochastic Extension

Question 2 extends the deterministic model of Question 1 to account for uncertainty in robot battery ranges. Each of the 100 scenarios $s \in \mathcal{S} = \{1, \ldots, 100\}$ in `range_scenarios.csv` represents one possible day, providing a realised range r_i^s for every robot $i \in \mathcal{I}$. Critically, not every robot needs to charge on every day: a robot with a high remaining range is unlikely to require charging, while one with a very low range almost certainly will. This uncertain demand is captured by a charging indicator $\delta_i^s \in \{0, 1\}$ determined by a Monte Carlo simulation.

Modelling Approach

We apply a two-stage stochastic programming framework with the Sample Average Approximation (SAA), in which the expected cost over $|\mathcal{S}| = 100$ scenarios is minimized:

Stage 1 **Here-and-now decisions** - made before any scenario is revealed. The station open indicators z_j, charger counts n_j, and station locations (X_j, Y_j) are optimized over the full distribution of scenarios. These decisions are fixed across all scenarios.

Stage 2 **Wait-and-see decisions** - made after scenario s is revealed. For each scenario, we determine which robots require charging (δ_i^s) and assign them to stations via drone (a_{ij}^s) or human transport (h_{ij}^s).

Generating the Charging Indicator δ_i^s

For each robot $i \in \mathcal{I}$ and scenario $s \in \mathcal{S}$, the charging probability is

$$p_i^s = \exp\!\big(-\lambda^2 (r_i^s - r_{\min})^2\big), \tag{21}$$

where r_i^s is the realized range in scenario s from `range_scenarios.csv`, $\lambda = 0.012$, and $r_{\min} = 10\,\text{km}$. A robot with range close to $r_{\min}$ charges with probability near 1; one with range close to $r_{\max} = 175\,\text{km}$ has probability near 0.

We adopt the Monte Carlo procedure of (Ljósheim et al., 2026) to determines whether robot i actually charges in scenario s. We draw $\psi_i^s \sim U(0, 1)$ with a fixed random seed for reproducibility, then

$$\delta_i^s = \mathbf{1}[\psi_i^s \leq p_i^s] \tag{22}$$

The set of robots requiring charging in scenario s is $\mathcal{I}^s = \{i \in \mathcal{I} : \delta_i^s = 1\}$.

Objective Function

The stochastic objective averages the Stage 2 assignment costs over all scenarios, with the Stage 1 infrastructure costs paid once:

$$f^{\text{RP}} = c_b \sum_{j \in \mathcal{J}} z_j + c_m \sum_{j \in \mathcal{J}} n_j + \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \left[c_c \sum_{i \in \mathcal{I}^s} \sum_{j \in \mathcal{J}} a_{ij}^s \big(d_{ij} + r_{\max} - r_i^s\big) + c_h \sum_{i \in \mathcal{I}^s} \sum_{j \in \mathcal{J}} h_{ij}^s \right] \tag{23}$$

The charging cost term $c_c(d_{ij} + r_{\max} - r_i^s)$ uses the scenario-specific range r_i^s since the robot charges from its current range back to $r_{\max}$. Human transport cost c_h is scenario-independent (flat cost regardless of distance).

Complete Stochastic Model

$$\min \quad c_b \sum_{j\in\mathcal{J}} z_j + c_m \sum_{j\in\mathcal{J}} n_j + \frac{1}{|\mathcal{S}|} \sum_{s\in\mathcal{S}} \Big[c_c \sum_{i\in\mathcal{I}^s} \sum_{j\in\mathcal{J}} a_{ij}^s (d_{ij} + r_{\max} - r_i^s) + c_h \sum_{i\in\mathcal{I}^s} \sum_{j\in\mathcal{J}} h_{ij}^s \Big]$$

$$\begin{array}{lll}
\text{s.t.} & \sum_{j\in\mathcal{J}} (a_{ij}^s + h_{ij}^s) = 1 & \forall\, i \in \mathcal{I}^s,\ s\in\mathcal{S} \\
& a_{ij}^s + h_{ij}^s \le z_j & \forall\, i \in \mathcal{I}^s,\ j\in\mathcal{J},\ s\in\mathcal{S} \\
& \sum_{i\in\mathcal{I}^s} (a_{ij}^s + h_{ij}^s) \le q \cdot n_j & \forall\, j\in\mathcal{J},\ s\in\mathcal{S} \\
& n_j \le m \cdot z_j & \forall\, j\in\mathcal{J} \\
& n_j \ge z_j & \forall\, j\in\mathcal{J} \\
& d_{ij}^2 = (x_i - X_j)^2 + (y_i - Y_j)^2 & \forall\, i\in\mathcal{I},\ j\in\mathcal{J} \\
& d_{ij} \le r_i^s + M(1 - a_{ij}^s) & \forall\, i \in \mathcal{I}^s,\ j\in\mathcal{J},\ s\in\mathcal{S} \\
& X_j,\ Y_j \in \mathbb{R} & \forall\, j\in\mathcal{J} \\
& n_j \in \{0,\dots,m\} & \forall\, j\in\mathcal{J} \\
& z_j,\ a_{ij}^s,\ h_{ij}^s \in \{0,1\} & \forall\, i \in \mathcal{I}^s,\ j\in\mathcal{J},\ s\in\mathcal{S} \\
& d_{ij} \ge 0 & \forall\, i\in\mathcal{I},\ j\in\mathcal{J}
\end{array}$$

where $\mathcal{I}^s = \{i \in \mathcal{I} : \delta_i^s = 1\}$ is the set of robots requiring charging in scenario s, and r_i^s is the realized range of robot i in scenario s. The Stage 1 variables (z_j, n_j, X_j, Y_j) are fixed across all scenarios; the Stage 2 variables (a_{ij}^s, h_{ij}^s) are scenario-specific.

Solution Method

We solve the Recourse Problem (RP) using the location–allocation heuristic from Question 1c, adapted for the stochastic setting:

1. **Initialize** station locations using k-means on all 1 072 robot coordinates (same as Q1b, $\gamma = 2$).
2. **Solve linearized ILP** with fixed station locations. Since (X_j, Y_j) are fixed, all distances d_{ij} are parameters and the problem reduces to a pure integer linear programme over all scenarios simultaneously. The ILP minimizes (23) subject to: each robot in $\mathcal{I}^s$ assigned exactly once per scenario (C1), only to open stations (C2), within capacity $q \cdot n_j$ per scenario (C3), and with charger bounds (C4).
3. **Relocate stations** via geometric median of robots most frequently assigned (across scenarios) to each station, using the range-safe Weiszfeld algorithm from Q1c.
4. **Repeat** until no improvement exceeds $\epsilon = £100$.

Training and test split: We use 80 of the 100 scenarios as training scenarios to solve the RP, and hold out 20 scenarios for out-of-sample validation. This tests whether the stochastic solution generalizes to unseen demand realizations.

Value of the Stochastic Solution (VSS)

The VSS quantifies the benefit of stochastic planning over naïve use of the deterministic Q1c solution. We define:

RP The Recourse Problem cost: solution optimized over the stochastic scenarios (23).

EEV Expected cost of the EV solution: we take the Q1c deterministic station layout (locations and charger counts fixed), evaluate it on the 100 stochastic scenarios by re-solving the Stage 2 assignment only.

$$\mathrm{VSS} = f^{\mathrm{EEV}} - f^{\mathrm{RP}} \geq 0. \tag{24}$$

A positive VSS means that optimizing over scenarios produces a cheaper expected cost than simply using the deterministic solution. As a percentage:

$$\mathrm{VSS}\,(\%) = \frac{f^{\mathrm{EEV}} - f^{\mathrm{RP}}}{f^{\mathrm{EEV}}} \times 100\%. \tag{25}$$

Computational Results

Table 5 reports the RP, EEV, and VSS on both the training and out-of-sample test scenarios.

Table 5: Stochastic solution results: RP, EEV, and VSS ($|\mathcal{S}| = 100$ scenarios, 80 training / 20 test).

Metric	Training (80 scenarios)	Test (20 scenarios)
Mean charging probability $\bar{p}$	0.5365	
Mean robots charging per scenario	573.5 / 1,072	
RP open stations	38	
RP total chargers	302	
RP cost – stochastic solution f^{RP} (£)	815,183.12	811,279.59
EEV cost – deterministic Q1c on scenarios (£)	1,108,136.70	1,105,582.22
VSS = $f^{\mathrm{EEV}} - f^{\mathrm{RP}}$ (£)	292,953.58	294,302.63
VSS (%)	26.44	26.62
RP solve time (s)	2,558.56	

The results yield several important insights about the value of stochastic planning for the Antarctic robot charging problem.

Fewer robots charge per scenario than assumed in Q1: The mean charging probability across all robots and scenarios is $\bar{p} = 0.536$, meaning approximately 574 of the 1 072 robots require charging on an average day. This is substantially fewer than the 1 072 assumed in the deterministic Q1 model, where every robot was charged every day. The stochastic model correctly captures this reduced demand.

The stochastic solution requires far fewer stations: The RP solution opens only 38 stations with 302 chargers, compared to 72 stations and 538 chargers in the Q1c deterministic solution. This reduction of 34 stations saves $34 \times £5{,}000 = £170{,}000$ in annual build costs alone. The stochastic solution places stations strategically near robots with high charging probability (low r_i^s), rather than building infrastructure to serve all robots simultaneously.

The VSS is large: 26.44% on training scenarios: VSS = £292,954, representing a 26.44% cost reduction from stochastic planning relative to deploying the deterministic solution. This is a substantial benefit, the deterministic Q1c solution over-builds because it assumes all robots charge every day. When evaluated on the stochastic scenarios (where only ∼574 robots charge per day), this over-building incurs unnecessary build and maintenance costs without proportional savings in assignment costs.

The solution generalizes well out-of-sample: On the 20 held-out test scenarios, the RP cost is £811,280 and the EEV cost is £1,105,582, giving $\text{VSS}_{\text{test}} = £294{,}303$ (26.62%). The out-of-sample VSS is essentially identical to the training VSS (26.44% vs 26.62%), confirming that the stochastic solution is robust and has not over-fitted to the 80 training scenarios. This is consistent with the findings of (Ljósheim et al., 2026), who reported that solutions become more robust as the number of scenarios increases.

The RP converges in 5 iterations: The location–allocation heuristic improves the objective from £1,108,137 (Q1b starting point) to £815,183 over 5 iterations (Figure 5), a 26.4% reduction. Most improvement occurs in the first iteration (£289,417 gain), with subsequent iterations making progressively smaller refinements as seen in Figure 6. The algorithm terminates when the improvement falls below the $\epsilon = £100$ threshold.

Comparison with Q1c deterministic solution: Table 6 summarizes the key differences between the Q1c and Q2 solutions.

Table 6: Comparison of Q1c deterministic and Q2 stochastic solutions (full 1,072-robot instance).

	Q1c (deterministic)	Q2 (stochastic RP)
Robots assumed to charge	1,072 (all)	∼574 (avg)
Open stations	72	38
Total chargers	536	302
Annual build cost (£)	360,000	190,000
Annual maintenance cost (£)	268,000	151,000
Avg charge cost/day (£)	23,629	5,562
Avg human transport/day (£)	654,000	483,575
Total annual cost (£)	1,305,629	815,183
VSS vs EEV (£)	—	292,954
VSS vs EEV (%)	—	26.44

$\text{VSS} = f^{\text{EEV}} - f^{\text{RP}}$, where $f^{\text{EEV}} = £1{,}108{,}137$ is the cost of the Q1b construction heuristic solution evaluated on the 80 training scenarios.

The stochastic solution achieves substantial savings across all cost components. Human transport costs fall by 26% (£483,575 vs £654,000) because the stochastic model places stations near robots that genuinely need charging on average, rather than trying to reach all 1,072 robots simultaneously. Build and maintenance costs are reduced by approximately 46% ($£190{,}000 + £151{,}000$ vs $£360{,}000 + £268{,}000$) due to the smaller number of stations required. Charging costs fall most sharply by 76% (£5,562 vs £23,629), reflecting the more targeted placement of stations close to high-probability robots, enabling more drone assignments at shorter distances. These results confirm that explicitly modelling stochastic demand is critical for cost-efficient charging infrastructure design..

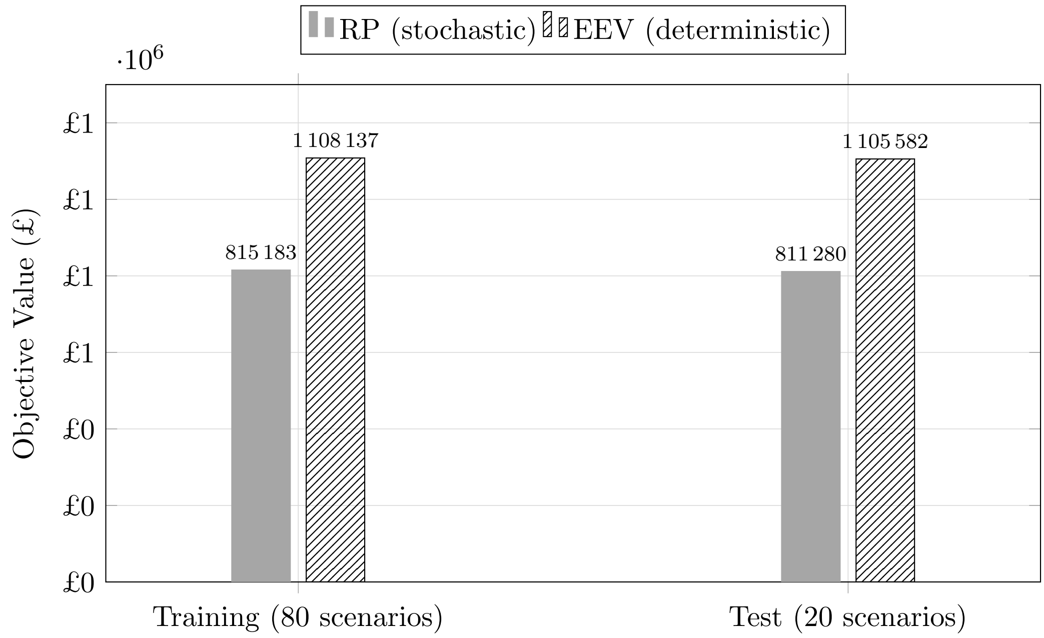


Figure 5: RP and EEV costs on training and test scenarios. The stochastic solution (RP) consistently outperforms the deterministic Q1c solution (EEV), saving over £292,000 (26.4%) on training scenarios and £294,000 (26.6%) on held-out test scenarios.

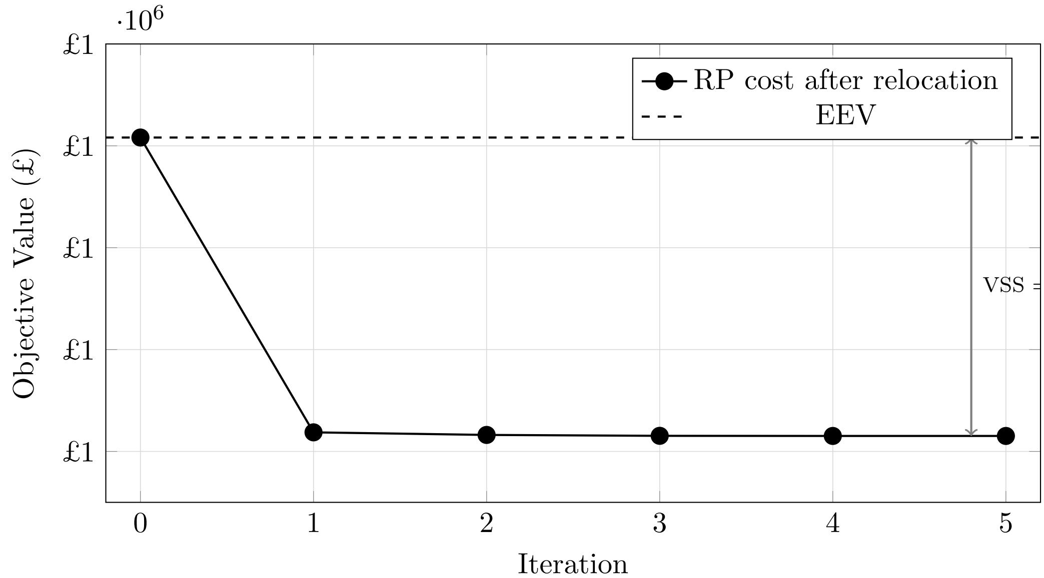


Figure 6: Convergence of the RP location–allocation heuristic over 5 iterations. The largest improvement occurs in iteration 1 (£289,417 gain) as the ILP re-optimises all assignments simultaneously. The dashed line shows the EEV baseline cost; the bracket indicates the VSS of £292,954 (26.4%).

6 Robot Allocation Plots

Results before improvement (Greedy assignment) | **Results after improvement(Local search)**

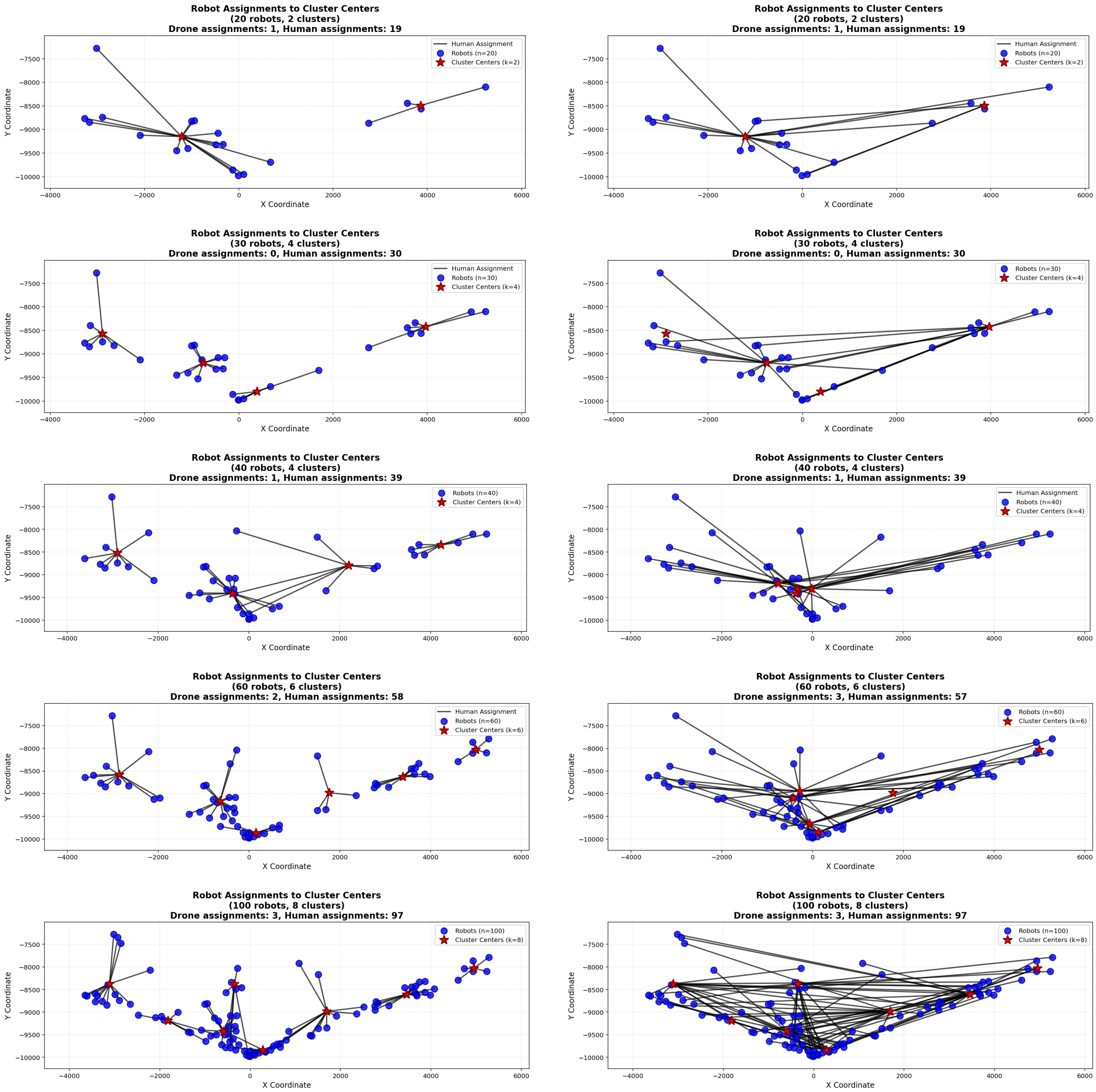


Figure 7: Robot assignment comparison: construction heuristic vs. improvement

References

Ljósheim, H., Jenkins, S., Searle, K., and Wolff, J. (2026). Optimal placement of electric vehicle slow-charging stations: A continuous facility location problem under uncertainty. *Computers & Operations Research*, 185:107289.